CASE

Marcus, 47 · pickup basketball · sudden "pop" at the back of the ankle · cannot rise on tiptoes · palpable gap above the heel · positive Thompson test · **complete Achilles tendon rupture**

BIO 004 · WEEK 2 · CASE DEEP DIVE

Fascicle Arrangement & Lever Systems

A weekend basketball game, a sound like a gunshot, and a calf that can no longer push off.

PATIENT CASE

Marcus is 47, plays pickup basketball "once a week, sometimes." He went up for a rebound and as he landed felt a **sudden sharp pain at the back of his right ankle and heard a pop**, "like someone shot a starter pistol." He fell. He thought someone had kicked him from behind.

In the ED he cannot **rise on tiptoes on the right leg**. There is a **palpable gap about 3 cm above the heel**. The **Thompson test is positive** (squeezing the calf does not produce plantar flexion of the foot). Imaging confirms a **complete rupture of the Achilles tendon**.

The Achilles is the largest, strongest tendon in the body, and it rides at the end of a beautifully arranged muscle group with a specific fascicle architecture that makes it powerful but vulnerable. To understand why it ruptures in the middle-aged weekend athlete, you have to understand how muscle fibers are organized within a muscle, how that arrangement trades off force for speed, and how the resulting force becomes movement through a class-2 lever at the foot.

> GUIDING QUESTION
>
> How are muscle fibers organized inside a muscle, why does the arrangement matter for force versus speed, and how does the foot use a class-2 lever to lift the entire body off the ground?

PREWORK

Companion to your Week 2 Muscle prework on fascicle arrangements and lever systems. Terms, video, and spaced recall live there.

1 FASCICLE BASICS

How a muscle is wired inside

A muscle's **fascicles** are bundles of muscle fibers wrapped in perimysium. The way fascicles are arranged inside the whole muscle is called the muscle's **fascicle architecture**. Two variables matter most: the *length* of fascicles relative to the muscle, and the *angle* at which they attach to the tendon.

The fundamental tradeoff is **force versus range/speed**. More fascicles in parallel (short, many) = more force, less range. Longer fascicles in series (long, fewer) = more range and speed, less force. Pennate arrangements maximize force per unit volume; parallel arrangements maximize range and shortening velocity.

2 FIVE ARRANGEMENTS

The shapes muscles come in

Parallel (fusiform)
Fascicles run parallel to the long axis. Spindle-shaped (fusiform: biceps brachii). Maximizes range and shortening velocity.

Circular (sphincter)
Fascicles arranged in concentric rings around an orifice. Orbicularis oris, orbicularis oculi, anal sphincter.

Convergent
Fascicles converge from a broad origin to a narrow insertion. Triangular shape: pectoralis major.

Pennate (unipennate)
Fascicles attach to one side of the tendon at an angle, like one half of a feather. Extensor digitorum longus.

Bipennate
Fascicles attach to both sides of a central tendon. Rectus femoris.

Multipennate
Multiple tendon branches each receive pennate fascicle attachments. Deltoid. Highest force-per-volume but smallest excursion.

Marcus's calf muscles are largely **pennate**: short fibers at an angle to the tendon, maximizing force production for the work of pushing off the ground. All that force is funneled into one tendon — the Achilles — at the heel.

3 THREE CALF MUSCLES, ONE TENDON

The triceps surae and the Achilles

The Achilles tendon (calcaneal tendon) is the common insertion of three muscles, called collectively the **triceps surae**:

Gastrocnemius (medial and lateral heads)
Originates above the knee on the femoral condyles. Crosses both knee and ankle. Pennate. Provides explosive plantar flexion. Active in running and jumping.

Soleus
Deep to gastrocnemius. Originates from tibia and fibula. Pennate. Largely Type I (slow oxidative) fibers. Active in standing and walking.

Plantaris (vestigial in some)
Long thin tendon. Small contribution to plantar flexion.

All three tendons fuse to form the Achilles, which inserts on the **posterior calcaneus**. The combined force of the triceps surae through that single tendon is what lifts the entire body weight onto the toes during push-off.

4 LEVERS

Force, fulcrum, and load

Every joint-tendon-bone system in the body acts as a lever. Three classes:

CLASS	ARRANGEMENT	TRADE-OFF	EXAMPLE
Class 1	Fulcrum between effort and load	can favor either	head nodding (atlanto-occipital joint)
Class 2	Load between effort and fulcrum	force advantage (effort moves less but lifts more)	standing on tiptoes (ball of foot = fulcrum; body weight = load; calf pull = effort)
Class 3	Effort between fulcrum and load	speed/range advantage (small force moves load far and fast)	most muscles in the body (biceps lifting a weight)

Class-3 levers (most of our muscles) sacrifice mechanical advantage for speed and range of motion. The biceps lifts a weight in your hand only a small distance with a lot of muscle contraction, but does it fast and over a large arc. Class-2 levers (the calf at the foot) sacrifice range for mechanical advantage. The calf shortens a few centimeters but lifts the body weight off the ground.

5 FORCE-LENGTH & MOMENT ARM

Why a tendon's position matters as much as its size

Force generation also depends on where the tendon attaches relative to the joint axis (the **moment arm**) and on how stretched or shortened the sarcomeres are (the **force-length relationship**).

The Achilles tendon's moment arm at the ankle is short — about 4-5 cm from the joint axis — which means the calf must generate a very large force to lift the body. The body weight times the distance from the ankle to the body's center of mass is the resistance the calf has to overcome. The math typically works out to the calf needing to generate *roughly two to three times body weight* for normal heel raise.

> **Why the middle-aged weekend athlete.** Tendons degenerate gradually with age: collagen disorganization, reduced vascularity, decreased elasticity. The Achilles has a notoriously poor blood supply in a zone 2-6 cm above its insertion. Decades of accumulated microdamage in that zone combined with a sudden eccentric load (landing from a jump) is the textbook rupture mechanism. Marcus probably loaded a partially degenerated tendon past its breaking point.

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6 WHAT THE THOMPSON TEST TESTS

Bedside anatomy in one squeeze

The **Thompson test**: patient lies prone with feet over the edge of the table. Examiner squeezes the calf. A normal, intact Achilles tendon transmits the squeeze to the calcaneus and the foot plantar-flexes. A ruptured Achilles cannot transmit the squeeze; the foot does not move.

The test works because of the anatomy. The calf muscles are connected to the calcaneus via the Achilles tendon. Compress the calf and the muscles bulge against the tendon, pulling the heel up. Cut that chain anywhere along the way and the signal stops. Positive Thompson is highly specific for complete rupture.

✎ **DRAW ZONE 1** -- *Draw the five fascicle arrangements (parallel, convergent, circular, pennate, multipennate) with one muscle example each.*

Use Fine pencil for the individual fascicles. Show the angle clearly in the pennate arrangements.

✎ **DRAW ZONE 2** -- *Sketch Marcus's foot from the side as a class-2 lever. Mark the fulcrum (ball of foot), the load (body weight at center of mass), and the effort (calf pulling via Achilles). Then draw where the Achilles ruptured.*

Mark the typical 2-6 cm avascular zone above the calcaneal insertion. That's where Marcus's tendon failed.

RETURNING TO MARCUS

Pennate calf, class-2 lever, degenerated tendon

Marcus's rupture combines anatomy at three scales: a pennate muscle architecture optimized for force, a class-2 lever optimized for advantage, and a single tendon at a single weak spot.

"*Sound like a gunshot*"

Complete tendon ruptures release stored elastic energy suddenly. The audible pop is the tendon snapping under tension. Reported in 80-90% of acute Achilles ruptures.

"*Cannot rise on tiptoes*"

The class-2 lever requires the calf-to-Achilles-to-calcaneus chain intact. Break the chain and the lever doesn't work. Body weight wins.

"*Palpable gap above the heel*"

Without continuous tendon, the muscle belly retracts upward and the distal stump pulls toward the calcaneus. The gap is the missing tendon.

"*Positive Thompson test*"

Squeezing the calf cannot generate plantar flexion without a continuous tendon. Specific for full rupture.

"*Why the middle-aged weekend warrior*"

Degenerated tendon (decades of microdamage in the avascular zone) + sudden eccentric load (jump landing) = rupture. Younger athletes have stronger tendons; older sedentary patients don't generate the load.

"*Why surgery vs casting*"

Modern care often opts for surgical repair in active patients for lower re-rupture rate and faster return to function. Non-operative care with functional bracing works well for less active or higher-surgical-risk patients. Both work; choice depends on patient goals.

Pennate, lever, moment arm, avascular zone. Marcus's ankle is biomechanics demonstrated at the bedside. When you go back to the prework, the lever diagrams will land differently because one of them just snapped on a basketball court.